Project Progress Report — Where We Stand

Project MAESTRO — Design and Evaluation of a Safe Multi-Agent System for Natural Language-Driven Desktop Task Automation **Group 298** · Shashank Gupta (A2345923073) · Seenu (A2345923074) · Jairaj Berry (A2345923013) **Guide:** Dr. Rajni Sehgal Kaushik · **Area:** Agentic AI with specialization in NLP **Institution:** Amity School of Engineering & Technology · B.Tech 7CSE (Evening), Session 2023–27 **Report date:** 19 August 2026 · **Current roadmap position:** Week 5 of 12 (Minor)

1. Executive Summary

Four weeks in, the project is **on schedule on the graded (Minor) track and ahead of schedule on the build (Major shadow) track**.

- **Minor (research track):** Title finalized, synopsis submitted, literature review Part-I written, a comparative study of existing agent frameworks completed, and the research-gap chapter drafted with a finalized novelty statement. This corresponds exactly to PERT Phases 1–2 (Weeks 1–5) being complete on time.
- **Major (shadow build track):** The safety-first core is not just designed — it is **coded and tested**. MAESTRO v0.2 exists: a working English → plan → safety gate → execution pipeline with **40 passing automated tests**, including adversarial tests (path-traversal escape, a "lying planner," and audit-log tampering).

The single most important status fact: **the research contribution (the deterministic safety layer) already runs on a real machine.** Everything remaining in the Minor semester is specification, evaluation design, and report writing *around* a system that demonstrably works.

2. What the WPRs Record (Weeks 1–4)

WPR	ROADMAP WEEK	WORK RECORDED	REPORT CHAPTER PRODUCED
WPR-1	Week 1	Project title, objectives, and scope finalized; research domains identified (AI, NLP, Multi-Agent Systems, Desktop Automation, AI Safety); synopsis submitted; feasibility stated (₹0 bill of material — free/open-source stack)	— (5% experimental work)
WPR-2	Week 2–3	Studied AI agents, NLP, and Large Language Models; reviewed AI-agent concepts; collected research papers; studied LLM architecture	**Literature Review (Part-I)**
WPR-3	Week 4	Studied existing AI desktop-automation frameworks — OpenClaw, Hermes, Open Interpreter — and compared their architectures and capabilities	**Comparative Study section**
WPR-4	Week 4–5	Research-gap analysis conducted; shortcomings identified in **transparency, safety, and modular task execution**; project contribution defined; novelty statement finalized	**Research Gap chapter**

The WPR trail already tells a clean research story: *domain identified → literature surveyed → existing systems compared → gap isolated → contribution stated.* That narrative arc is exactly what the final report's Chapters 1–3 need.

3. Research Done So Far (Track A — the Minor's substance)

3.1 Literature survey (in progress, on schedule)

The survey clusters papers into six threads: computer-use agents, task planning & decomposition, agent safety & guardrails, prompt injection, NL→structured-action datasets, and agent benchmarks. Seed systems mined for citations: **Agent S / Agent S2, OSWorld, TPTU, Voyager**, plus the indirect-prompt-injection literature.

3.2 Comparative study of existing systems (WPR-3)

What the comparison of Open Interpreter–class systems (and the OpenClaw/Hermes family of open agent frameworks) established:

PROPERTY	EXISTING DESKTOP/LLM AGENTS	CONSEQUENCE
Capability (open-ended NL)	Strong and improving	Capability is **not** the gap
Action representation	Generated code or free-form tool calls	Unanalyzable before execution; unsafe by construction
Risk classification	Absent	No principled notion of "dangerous action"
Preview / dry-run	Absent or cosmetic	User approves blind
Prompt-injection defense	Absent	File/web content can hijack the agent
Undo / reversibility	Absent	One wrong plan = permanent loss
Audit trail	Plain logs at best	No tamper evidence, no accountability
Safety **metrics**	Essentially never reported	Safety cannot even be compared across systems

3.3 The research gap (WPR-4 — the novelty statement)

The four-part defensible gap, each part observable in the literature matrix:

1. **Prior work measures capability, not safety.** There is no accepted safety metric for desktop agents.
2. **Confirmation dialogs are UI; ours is architecture.** A typed intermediate representation (Action IR) with deterministic risk scoring and dry-run has *testable properties*; a popup does not.
3. **No existing desktop-agent work evaluates prompt-injection resistance as a first-class metric.**
4. **No public NL→plan dataset exists** for cross-platform desktop automation with safety annotations. (Ours: **DeskPlan**, 3,000+ pairs target.)

The identified shortcomings — **transparency, safety, modular task execution** — map one-to-one onto MAESTRO's three pillars: the dry-run preview + audit log (transparency), the deterministic risk/consent engine (safety), and the closed verb registry + per-OS executor layer (modularity).

---

4. Build Progress (Track B — the Major's head start)

Not graded this semester, but decisive for the 8th semester. Verified today (19 Aug 2026): **40/40 tests pass** in `Major/`.

v0.1 — the safety-first core (no LLM, deliberately)

MODULE	WHAT IT DOES
`maestro/ir/`	**Action IR**: typed plans, DAG validation, `$var` dataflow between actions
`maestro/registry.py`	**Closed verb registry** — an unknown verb is a rejected plan

MODULE	WHAT IT DOES
`maestro/safety/paths.py`	Path allow/denylist with canonicalization (traversal- and symlink-safe)
`maestro/safety/scorer.py`	**Deterministic risk scorer** R0–R3 — monotonic, fail-closed, no LLM involved
`maestro/safety/audit.py`	Hash-chained, tamper-evident audit log
`maestro/executor/fs.py`	7 portable file verbs, each with `dry_run()` and `undo()`
`maestro/orchestrator.py`	dry-run → consent gate → topological execution → rollback

v0.2 — the planner and the learning loop

- `maestro ask "<instruction>"` — the full pipeline: English → local LLM (Ollama, **constrained JSON decoding** with the verb enum = the closed registry) → IR validation → deterministic risk scoring → dry-run preview → consent → execution → **episode recorded**.
- `maestro learn` — episode statistics + export of JSONL training candidates for the future LoRA fine-tune.
- Planner schema hardened to force explicit dataflow (`$var` bindings) — the model cannot smuggle implicit state between steps.

The three properties the test suite pins down (evidence, not claims)

1. **A lying planner cannot downgrade risk** — the planner's `risk_hint` is recorded for a hint-agreement metric but *ignored* for decisions.
2. **Path escapes fail closed** — `workspace/../secrets/x` and symlinks into denied directories are DENIED before matching.
3. **Editing the audit log is detectable** — the hash chain breaks.

Known issue, documented deliberately

A refused unsafe instruction currently exports as a "completed" episode. Before any training run, every candidate must be human-labeled with `expected_behavior` (execute / clarify / refuse) so **refusals teach refusal, not compliance**. This human-verification step is recorded as non-optional in the dataset design.

---

5. Progress vs. the PERT Schedule

PERT PHASE	WEEKS	STATUS	EVIDENCE
1. Literature Review	1–3	🟢 ~80% (finishing to 30 papers)	WPR-2; Literature Review Part-I
2. Existing System Study & Gap Analysis	4–5	🟢 **Complete**	WPR-3, WPR-4; gap chapter drafted
3. Requirement Analysis	6	🟡 **In progress (current week)**	PRD exists (FR-01…FR-62, NFR-01…NFR-10); being formalized into the report chapter
4. System Architecture Design	7–8	🟡 Design done, diagrams pending	Full architecture doc (L0–L6, Action IR, trust model); 10 graded diagrams to produce
5. Module Development	9–14	🟢 **Ahead** — core modules already coded	MAESTRO v0.2, 40 tests
6. System Integration	15–16	🟢 Ahead — end-to-end `ask` pipeline works	CLI demo runs offline

PERT PHASE	WEEKS	STATUS	EVIDENCE
7. Testing & Evaluation	17–18	🟡 Methodology designed; harness pending	Metrics, baselines B0–B5, ablations A0–A8 specified
8. Documentation & Final Report	19–20	🔴 Starting now	This folder

Overall Minor progress: ~40% of the semester's graded deliverables; the underlying system is further along than the schedule requires.

Report-writing status to carry into WPR-5: Literature Review Part-I ✅ · Comparative Study ✅ · Research Gap ✅ · Requirements chapter ⏳ (this week).

---

6. What Remains in the Minor (Weeks 5–12)

1. **Week 5 (now):** Requirements chapter with traceability IDs; 8–10 full-form use cases; use-case diagram; feasibility study write-up.
2. **Week 6:** Freeze **Action IR v1.0** spec, risk taxonomy R0–R3, trust model T0–T2. **Scope freeze** — everything proposed later goes to Future Work.
3. **Weeks 7–8:** Architecture chapter (~15 pp); the 10 graded diagrams (C4, DFD, sequence, state machine, ER, deployment); algorithm pseudocode chapter.
4. **Week 9:** Evaluation methodology chapter — 100-task benchmark design, formal metric definitions, baselines, ablations, user-study protocol. *Ethics-clearance question for the user study must be answered by the guide — long lead time.*
5. **Week 10:** 300+ seed instruction→plan pairs; 50 adversarial cases; annotation guidelines; Cohen's κ on a 50-pair overlap.
6. **Weeks 11–12:** Report assembly, plagiarism check, guide review, demo video, viva prep, WPR-12.

---

7. How the Minor Feeds the Major

The Minor is the **research**; the Major is the **outcome**. Every Minor deliverable is an input the Major consumes directly:

MINOR PRODUCES (RESEARCH)	MAJOR CONSUMES IT TO PRODUCE (OUTCOME)
Literature matrix + gap analysis	The paper's related-work and claims
Action IR v1.0 + risk taxonomy specs	The frozen contract all 8th-sem code implements
Evaluation methodology (metrics, 100 tasks, ablations)	The benchmark run (3,000 runs) and results chapter
350 seed dataset pairs + guidelines + κ	Scaled to **DeskPlan 3,200** pairs → LoRA fine-tune
Working vertical slice (v0.2)	Windows backend, browser/app/system verbs, voice, GUI
12 WPRs	The audit trail of the project itself

Headline targets the Major must hit: **Unsafe Execution Rate = 0** on the adversarial suite · Injection Resistance ≥ 90% · Task Success ≥ 70% on P0 categories · fine-tuned 3B vs. frontier comparison · user study n ≥ 15 · public dataset release.

(All terms used above are defined in 02-Clarifications-and-Key-Terms.md.)